

Kyle Webb

Staff Data Scientist - statistics, data engineering, and applied AI

webb.kyle@outlook.com · github.com/webbkyle

PROFESSIONAL SUMMARY

Data scientist and data engineer with over ten years of experience spanning scientific research, federal health programs, and industry. Operates in both worlds: architects the data platforms, pipelines, and governance that make data trustworthy and readily usable, then applies machine learning and applied statistics to turn that data into predictions and decisions. Having built both sides, he knows how to serve data efficiently for modeling and predictive analytics - from raw ingestion to deployed model. Deep hands-on expertise across Databricks, AWS, and Terraform, paired with AI-augmented analytics and a statistician's rigor.

EXPERIENCE

eSimplicity - Staff Data Scientist

Sept 2023 - Present · Remote

Senior Data Scientist, Apr 2022 - Aug 2023 · AI Data Analyst Fellow, Rapid Innovation Center

- Led a team of six DevOps and data engineers delivering scheduled, validated data distributions across 20 CDR sources - terabytes of claims, provider, and beneficiary data on AWS MWAA, Terraform, and Databricks.
- Migrated all legacy EMR workloads to Databricks with automated validation, testing, and monitoring - cutting median runtime 82% and median cost 93%.
- Designed data-governance workflows enforcing strict role-based access controls for 2,000+ users in a Kinesis-driven data repository.
- As AI Data Analyst Fellow, piloted an AI-agent cost-analysis application for Databricks resources, with an MCP server letting users ask live questions about their cost data.
- Established Databricks Delta Shares for governed, read-only sharing across workspaces, reducing data duplication costs.

GovernmentCIO - Data Scientist

July 2021 - Apr 2022 · Remote

- Sole R/Python developer automating PowerPoint reporting on VA IT infrastructure; reported major incident data to Operations Triage Group leadership and presented a weekly report to 200+ attendees including the VA Chief Information Officer.
- Delivered querying, cleaning, and analysis for 30+ briefs on IT topics - from invalid security certificates to telecom and network outages - for the largest healthcare network in the U.S., some with one-day turnarounds.

National Human Genome Research Institute (NHGRI) - Bioinformatics Scientist

Oct 2020 - July 2021 · Bethesda, MD

- Built the lab's workspace in the N3C data enclave; completed data use requests and agreements for COVID-19 EHR ETL; coached teammates on PySpark and SparkR.
- Gathered and assessed EHR data from millions of patients for a COVID-19 Phenotype-Wide Association Study (PheWAS).
- Pulled, mapped, and aggregated N3C and All of Us enclave data for a participant comparison study, using stratified sampling across race, gender, age, and ethnicity.

National Institute of Allergy and Infectious Diseases (NIAID) - Data Science Fellow

June 2018 - Sept 2020 · Bethesda, MD

- Ran ETL and parallelized HPC machine-learning jobs over 4TB+ of Cerner health data for rare-disease classification.
- Improved computational speed and network visualization for SIGNAL, a public genetic screening tool (R Shiny + D3.js) - delivered in four months against a six-month plan.
- Scrubbed PII from 32,000 clinical notes, then trained and compared two NLP models for Human Phenotype Ontology assignment, with t-SNE hierarchical renderings of results.
- Extended the R package ihazr for semiparametric hazard regression estimation and visualization.
- Completed five data-science rotations across the institute - biostatistics, immunology, bioinformatics, grants analytics, epidemiology (detailed on CV).

EDUCATION

M.S., Statistics - Virginia Tech (2015 - 2016)

B.S., Statistics - Wake Forest University (2011 - 2015), GPA 3.4/4.0

CERTIFICATIONS

AWS Certified Cloud Practitioner (2023) · AWS Certified Solutions Architect - Associate (2026) · Databricks Certified Data Engineering Associate (2026)

TECHNICAL SKILLS

Python · R · SQL · Databricks · AWS (S3, EMR, Glue, MWAA, Kinesis) · Microsoft Azure · Terraform · RStudio · Git/Linux · JavaScript, D3.js